Astrolabe: Steering Forward-Process Reinforcement Learning for Distilled Autoregressive Video Models

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Research Questions

- How can we efficiently align distilled autoregressive video models with human preferences without expensive re-distillation or memory-intensive reverse-process optimization?
- How can reinforcement learning be scaled to long video generation while maintaining memory efficiency and temporal coherence?
- How can reward hacking be mitigated in multi-objective video generation alignment?

Core Idea

Distilled autoregressive video models are fast but produce videos misaligned with human preferences. Existing RL methods either require expensive re-distillation or memory-intensive reverse-process optimization that defeats the efficiency purpose. Astrolabe's key insight is using forward-process RL that contrasts positive/negative samples at inference endpoints only, avoiding trajectory storage. For long videos, it uses a rolling KV-cache with clip-level updates while preserving historical context, and prevents reward hacking via multi-reward objectives with uncertainty-aware selective regularization.

Key Formula

Implicit Policy Interpolation

$$v^+ = (1-\beta)v_{\theta_{\text{old}}} + \beta v_\theta, \quad v^- = (1+\beta)v_{\theta_{\text{old}}} - \beta v_\theta$$

v^+ and v^- are implicit positive/negative policies, v_θ is current velocity predictor, v_θ_old is old velocity predictor, β controls interpolation strength. Creates contrasting policies for forward-process RL without requiring trajectory storage.

Group-wise Advantage

$$A^{(i)} = R(x_n^{(i)}, c) - \frac{1}{G}\sum_{j=1}^{G} R(x_n^{(j)}, c)$$

A^(i) is advantage for candidate clip i, R is reward function, x_n^(i) is i-th candidate clip, G is group size, c is text condition. Computes clip-level advantages via group mean-centering for local policy improvement.

Key Figure

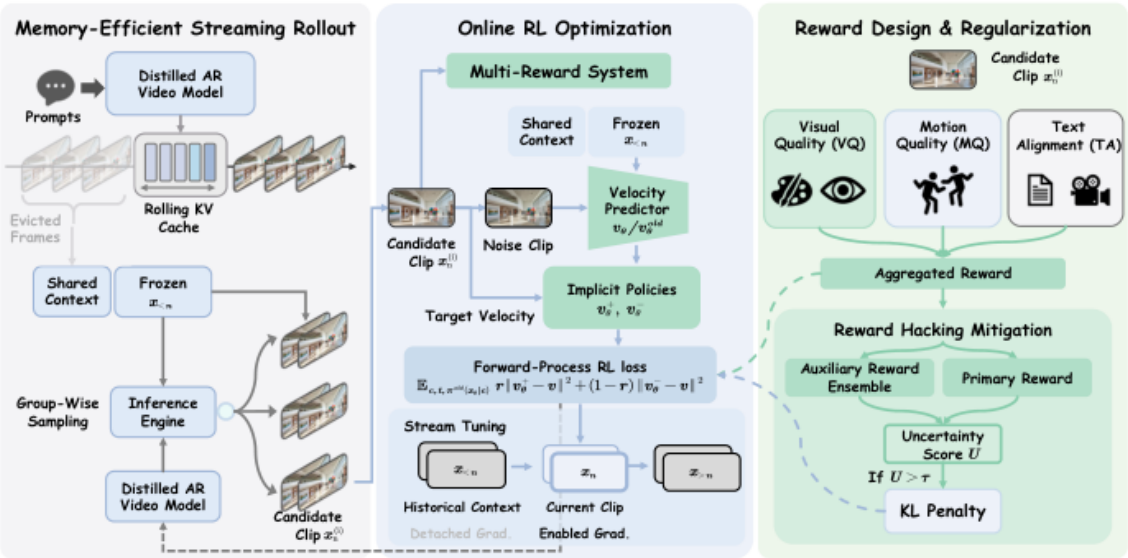

Fig. 2: Overview of Astrolabe. We propose a memory-efficient RL framework for distilled streaming video models. The method combines group-wise streaming rollout using a rolling KV cache for efficient group-wise sampling (see left), and clip-level forward-process RL for solver-agnostic optimization (see middle). To scale to long videos, we utilize Streaming Long Tuning with detached historical gradients. Furthermore, a multi-reward formulation paired with uncertainty-based selective regularization is employed to effectively mitigate reward hacking during training (see right). The pseudocode of the algorithm can be found in the supplementary materials.

mechanism for scalable exploration. Section 3.3 presents the online RL optimization strategy, encompassing clip-level forward-process RL and streaming long tuning. Finally, Section 3.4 formulates our multi-reward design and selective regularization approach to mitigate reward hacking.

3.1 Preliminaries

Autoregressive Video Diffusion Models. AR video model factorizes the joint distribution as $p(x_{1:N}) = \prod_{i=1}^{N} p(x_i|x_{<i})$. Following the flow matching formulation, each conditional $p(x_i|x_{<i})$ is modeled by defining a probability path $x_i^t = (1-t)x_i + t\epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, I)$ and $t \in [0, 1]$. The model predicts the velocity field v_θ conditioned on text c and the KV cache of preceding frames. Training paradigms such as Teacher Forcing (TF) and Diffusion Forcing (DF) minimize the frame-wise MSE between the predicted and true targets. In TF, timesteps t are shared across frames with clean ground-truth context $x_{<i}$, whereas in DF, independent timesteps t_i are sampled for each frame using noisy context $x_{j<i}^{t_j}$. Both suffer from exposure bias due to the mismatch between training context and inference-time generation. To mitigate this, Self-Forcing [22] employs autoregressive rollouts $\{x_{1:N}^{\theta}\} \sim \prod_{i=1}^{N} p_\theta(x_i|x_{<i})$ to simulate inference dynamics. The objective aligns the velocity predictions of the model on these self-generated trajectories with the scores provided by teacher model.

Figure 2: Shows the complete Astrolabe architecture including memory-efficient streaming rollout, forward-process RL optimization,

and multi-reward regularization system.

Limitations

- Limited to distilled models with T=4 timesteps, may not generalize to other distillation levels
- Rolling KV-cache with fixed frame sink size S may not capture all necessary long-range dependencies
- Multi-reward balancing requires careful hyperparameter tuning and may not generalize across different video domains
- Evaluation primarily on specific distilled AR baselines, broader applicability unclear